Factor Premia in the DAX

A point-in-time study of German large caps, benchmarked against the S&P 500

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Data through 2026-10-14 · Specification frozen 2026-10-14

Abstract

I test five well-documented price-based equity factors (momentum, short-term reversal, low volatility, low beta and the 52-week high) and an equal-weight composite inside the DAX, from 2006-02 to 2026-09 (248 months). The universe is rebuilt day by day from the 43 documented index changes, so every portfolio only holds stocks that were in the index at the time. The strongest signal is *Momentum 12-1*, whose long-short portfolio earns 7.2% per year after costs (Sharpe 0.40); its mean return is only marginally significant (10% level). After a Holm correction for testing six strategies, no long-short premium remains significant at the 5% level. Across all 36 specifications that were run, an unskilled researcher would expect a best Sharpe ratio of 0.43 by luck alone. No strategy has a deflated Sharpe ratio above 0.95, so none of them can be called a discovery on this data. The long-only composite portfolio returns 9.8% per year against 7.4% for the DAX, with a beta of 0.65, a tracking error of 9.7% and an information ratio of 0.16. Its CAPM alpha of 4.2% per year has a t-statistic of 2.80. Using today's members for the whole history raises the equal-weight return from 7.1% to 11.0% per year, a bias of 3.8% per year that would be mistaken for skill. All numbers in this paper are produced by code and update automatically when new data arrives.

1 Motivation

Factor studies on the DAX are easy to get wrong. The index has only 30 members until September 2021 and 40 afterwards, so a few stocks drive every result. Free data contains only today's survivors, which turns the past into a story of winners. Finally, with enough variants, something always "works". This study addresses each problem directly:

1. **Point-in-time universe.** Membership is reconstructed from the current constituent list and every inclusion and exclusion since 2005. Coverage gaps from delisted companies are measured and reported, not hidden (Section 2).
2. **Pre-specified design.** All factor definitions follow the original papers. Parameters are fixed in one configuration file and were not tuned; the specification was frozen on 2026-10-14.
3. **Honest inference.** Newey–West t-statistics, the probabilistic and deflated Sharpe ratio (Bailey and López de Prado, 2014), Holm-adjusted p-values and a robustness grid of 36 variants that all count as trials.
4. **Realistic implementation.** Trades happen one day after the signal, cost 10 bps per side, and short positions pay 25 bps p.a. in lending fees.
5. **Live record.** Everything after the freeze date is genuine out-of-sample data (Section 7).

2 Data and universe

Sources. Daily total-return prices (adjusted for dividends and splits) come from Yahoo Finance for 58 stocks. The benchmark is the DAX performance index, which reinvests dividends. The S&P 500 total-return index is converted into euros at the daily EUR/USD rate. The risk-free rate is the German 3-month interbank rate from FRED. Index changes are taken from the Wikipedia list of DAX changes and checked mechanically: the reconstructed index must contain exactly 30 members on every day before 20 September 2021 and 40 afterwards.

Coverage. 65 companies were DAX members at some point since 2005. For 7 of them no free price history exists (Table 1), mostly takeovers and insolvencies before 2013. On average 96.9% of the index is covered (Figure 1). The missing names are excluded from the universe while they were members. This is the remaining survivorship bias, and it is small but not zero: Wirecard's collapse in 2020, for example, is not in the data.

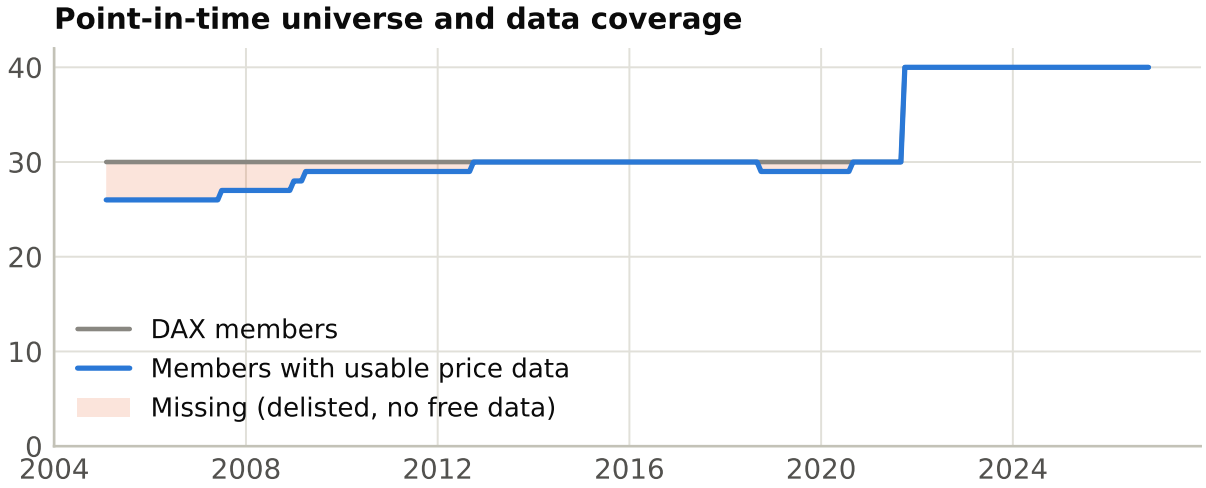


Figure 1: DAX members and members with usable price data. The shaded area is the gap left by delisted companies without free data.

Table 1: Index members without free price data

Company	In DAX from	Until
Altana	before 2005	2007-06-18
Deutsche Postbank	2006-09-18	2009-03-23
Hypo Real Estate	before 2005	2008-12-22
HypoVereinsbank	before 2005	2005-12-19
MAN	before 2005	2012-09-24
Schering	before 2005	2006-09-18
Wirecard	2018-09-24	2020-08-24

These stocks are excluded while they were members, so the effective universe is slightly smaller than the index in those periods.

Cleaning. Free data has defects. Each correction is applied explicitly and logged (Table 2). One share split was missing from the Yahoo series. On spin-off ex-dates Yahoo's adjustment for German stocks is inconsistent, so the parent's return on that day is replaced by the DAX return; this removes the error at the cost of one day of stock-specific information. Jumps of more than 20% that reverse the next day are treated as bad ticks. All other large moves are kept: 23 single-day moves above 25% remain in the sample after inspection, such as the Volkswagen short squeeze of October 2008 or the Siemens Energy profit warnings of 2023.

Survivorship bias. To show why the point-in-time universe matters, I compare an equal-weight portfolio of the actual members with one that simply uses today's members for the whole history (Figure 2). Using today's members for the whole history raises the equal-weight return from 7.1% to 11.0% per year, a bias of 3.8% per year that would be mistaken for skill.

Table 2: Data-quality corrections

Date	Stock	Fix	Raw	Clean	Reason
2005-01-31	Bayer	spinoff	−2.2%	1.3%	Spin-off of Lanxess (1 LXS per 10 BAYN)
2007-06-11	Deutsche Börse	split	−49.4%	1.2%	2-for-1 share split not reflected in the Yahoo series
2008-10-10	Deutsche Wohnen	bad tick	−20.1%	0.0%	jump reversed next day
2008-10-13	Deutsche Wohnen	bad tick	30.7%	0.0%	jump reversed next day
2013-07-08	Siemens	spinoff	4.1%	2.1%	Spin-off of Osram (1 OSR per 10 SIE)
2016-09-12	E.ON	spinoff	−6.0%	−1.3%	Spin-off of Uniper (1 UN01 per 10 EOAN)
2020-09-28	Siemens	spinoff	9.2%	3.2%	Spin-off of Siemens Energy (1 ENR per 2 SIE)
2021-09-16	Continental	spinoff	−6.2%	0.2%	Spin-off of Vitesco (1 VTSC per 5 CON)
2021-12-10	Mercedes-Benz Group	spinoff	3.6%	−0.1%	Spin-off of Daimler Truck (1 DTG per 2 DAI)

Raw = return in the Yahoo series, Clean = return used in the study.

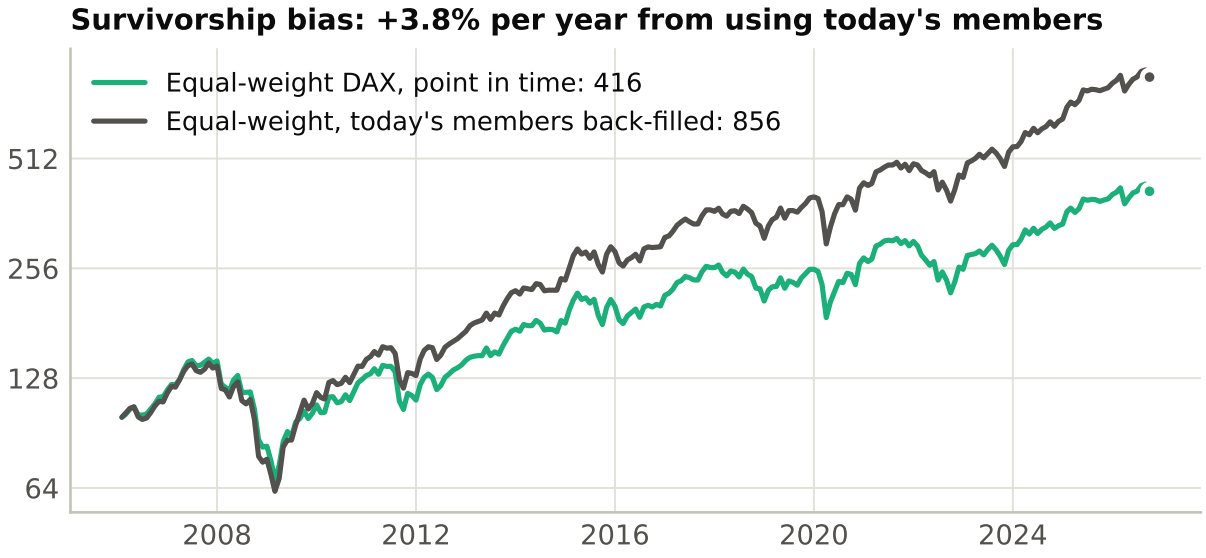


Figure 2: Equal-weight portfolios, monthly rebalanced, after costs. The grey line uses today's 40 members back-filled to 2006 and is therefore biased.

3 The market: DAX versus S&P 500

Before testing factors it helps to know the market they live in. Measured in euros, the S&P 500 compounded at 11.5% per year versus 7.4% for the DAX, with lower volatility (14.4% vs 17.7%). The correlation of monthly returns is 0.69 and the S&P 500's beta to the DAX is 0.56, so US equities diversified a euro investor's DAX risk rather than replicating it. Table 3 reports the full set of risk and return measures, and Figures 3 to 7 show the time series, rolling risk, calendar returns, the month-by-month history and the fat-tailed distribution of daily DAX returns.

Table 3: Benchmarks, 2006-02 to 2026-09

	DAX	S&P 500 (EUR)	Equal-weight PIT	Equal-weight survivors
CAGR	7.4%	11.5%	7.1%	11.0%
Volatility p.a.	17.7%	14.4%	17.5%	17.9%
Sharpe ratio	0.43	0.74	0.41	0.61
standard error	0.23	0.23	0.23	0.23
Sortino ratio	0.62	1.16	0.60	0.91
Maximum drawdown	−52.4%	−46.9%	−53.3%	−55.5%
Longest drawdown (months)	64	57	62	37
Calmar ratio	0.14	0.24	0.13	0.20
Skewness (monthly)	−0.49	−0.35	−0.57	−0.47
Excess kurtosis (monthly)	1.55	0.81	1.81	2.83
VaR 95% (monthly)	8.7%	6.8%	7.9%	7.4%
Expected shortfall 95%	11.9%	9.0%	11.9%	11.8%
Positive months	59.7%	64.1%	60.9%	61.3%
Best month	16.8%	14.4%	16.3%	22.2%
Worst month	−19.2%	−12.6%	−18.9%	−20.9%
Beta to DAX	–	0.56	0.97	0.98
Correlation with DAX	–	0.69	0.98	0.97

Monthly data. Sharpe ratios use the German 3-month rate; the standard error follows Mertens (2002), which allows for skewness and fat tails. VaR and expected shortfall are historical, one-month, 95%. "Equal-weight PIT" holds all point-in-time members with equal weights and pays the same trading costs as the strategies.

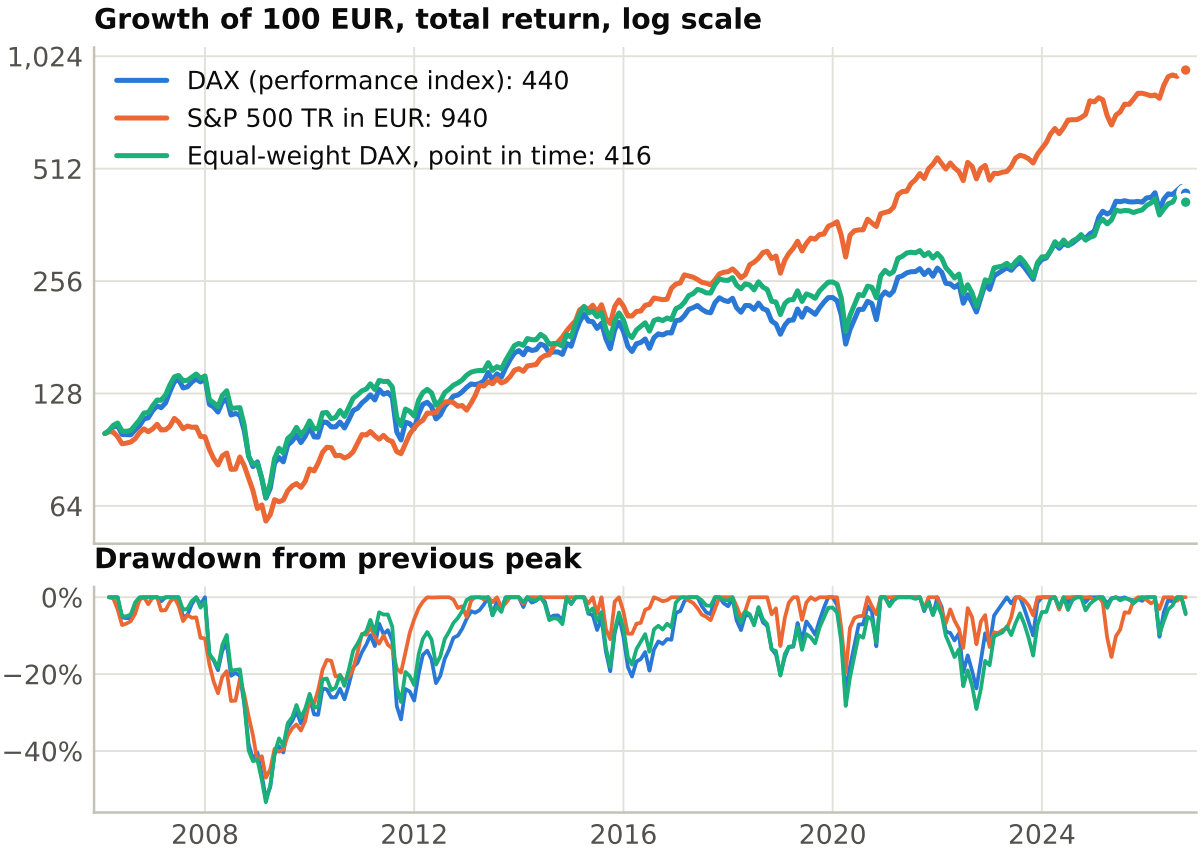


Figure 3: Total return in euros and drawdowns. Legend values are the final level of 100 invested at the start.

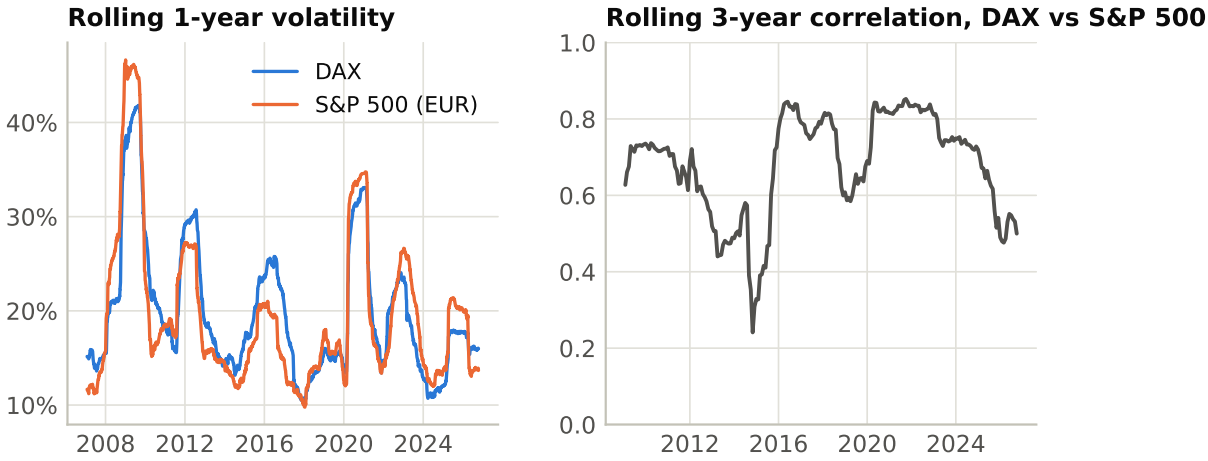


Figure 4: Rolling one-year volatility from daily returns and rolling three-year correlation of monthly returns.

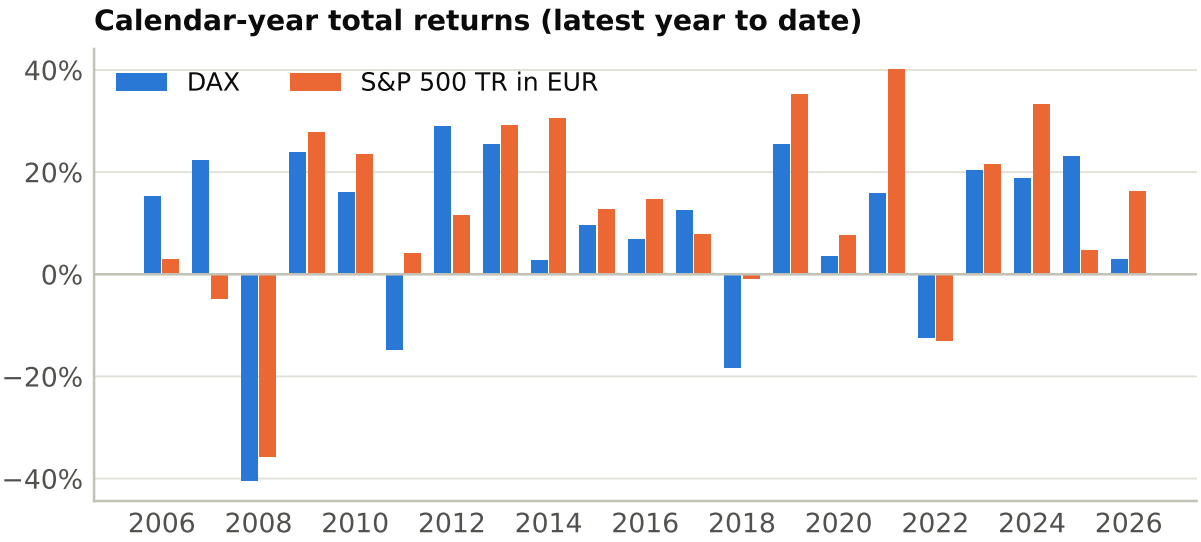


Figure 5: Calendar-year returns.

DAX monthly returns in %

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2006		1.2	3.0	0.7	-5.3	-0.2	-0.0	3.1	2.5	4.4	0.6	4.6
2007	2.9	-1.1	3.0	7.1	6.4	1.6	-5.3	0.7	2.9	2.0	-1.9	2.5
2008	-15.1	-1.5	-3.2	6.3	2.1	-9.6	1.0	-0.9	-9.2	-14.5	-6.4	3.0
2009	-9.8	-11.4	6.3	16.8	3.6	-2.7	10.9	2.5	3.9	-4.6	3.9	5.9
2010	-5.9	-0.2	9.9	-0.3	-2.8	0.0	3.1	-3.6	5.1	6.0	1.3	3.4
2011	2.4	2.8	-3.2	6.7	-2.9	1.1	-2.9	-19.2	-4.9	11.6	-0.9	-3.1
2012	9.5	6.1	1.3	-2.7	-7.3	2.4	5.5	2.9	3.5	0.6	2.0	2.8
2013	2.1	-0.4	0.7	1.5	5.5	-4.7	4.0	-2.1	6.1	5.1	4.1	1.6
2014	-2.6	4.1	-1.4	0.5	3.5	-1.1	-4.3	0.7	0.0	-1.6	7.0	-1.8
2015	9.1	6.6	5.0	-4.3	-0.4	-4.1	3.3	-9.3	-5.8	12.3	4.9	-5.6
2016	-8.8	-3.1	5.0	0.7	2.2	-5.7	6.8	2.5	-0.8	1.5	-0.2	7.9
2017	0.5	2.6	4.0	1.0	1.4	-2.3	-1.7	-0.5	6.4	3.1	-1.6	-0.8
2018	2.1	-5.7	-2.7	4.3	-0.1	-2.4	4.1	-3.4	-0.9	-6.5	-1.7	-6.2
2019	5.8	3.1	0.1	7.1	-5.0	5.7	-1.7	-2.0	4.1	3.5	2.9	0.1
2020	-2.0	-8.4	-16.4	9.3	6.7	6.2	0.0	5.1	-1.4	-9.4	15.0	3.2
2021	-2.1	2.6	8.9	0.8	1.9	0.7	0.1	1.9	-3.6	2.8	-3.8	5.2
2022	-2.6	-6.5	-0.3	-2.2	2.1	-11.2	5.5	-4.8	-5.6	9.4	8.6	-3.3
2023	8.7	1.6	1.7	1.9	-1.6	3.1	1.9	-3.0	-3.5	-3.7	9.5	3.3
2024	0.9	4.6	4.6	-3.0	3.2	-1.4	1.5	2.2	2.2	-1.3	2.9	1.4
2025	9.2	3.8	-1.7	1.5	6.7	-0.4	0.7	-0.7	-0.1	0.3	-0.5	2.7
2026	0.2	3.0	-10.3	7.1	3.3	-0.4	2.5	2.5	-4.0			

Figure 6: Every monthly DAX return of the sample.

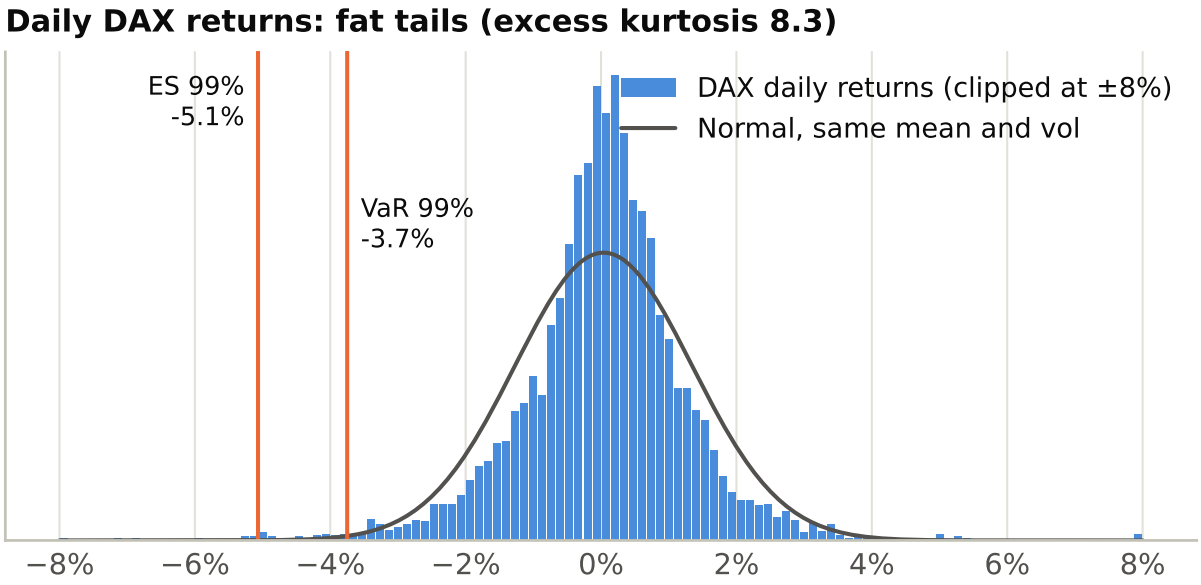


Figure 7: Daily DAX returns against a normal distribution with the same mean and volatility. VaR and expected shortfall at 99% are historical.

4 Methodology

Signals. Each signal is computed at the close of the last trading day of month t from daily total returns $r_{i,d}$ of the previous $L = 252$ trading days. A stock needs at least 200 valid returns in that window and must be a DAX member on the formation day. Higher values mean higher expected returns.

$$\begin{aligned}
\text{Momentum 12-1} &= \textstyle\prod_{d=t-251}^{t-21}(1+r_{i,d}) - 1 && \text{(Jegadeesh and Titman, 1993)}\\
\text{Short-term reversal} &= -\Big(\textstyle\prod_{d=t-20}^{t}(1+r_{i,d}) - 1\Big) && \text{(Jegadeesh, 1990)}\\
\text{Low volatility} &= -\sigma_i\sqrt{252} && \text{(Ang et al., 2006; Blitz and van Vliet, 2007)}\\
\text{Low beta} &= -\big(0.6\,\hat{\beta}_i + 0.4\big) && \text{(Frazzini and Pedersen, 2014)}\\
\text{52-week high} &= P_{i,t} \,/\, \max_{d\in(t-252,t]} P_{i,d} && \text{(George and Hwang, 2004)}
\end{aligned}$$

Here $\hat{\beta}_i = \operatorname{cov}(r_i, r_{\text{DAX}})/\operatorname{var}(r_{\text{DAX}})$ over the same window, shrunk towards 1. The composite is the average of the five cross-sectional z-scores, each winsorised at ± 3. Its weights are equal by construction and were not optimised.

Portfolios. Stocks are sorted into terciles each month (about 10 per leg with 30 members, 13 with 40). Portfolios are equal-weighted, bought at the close one trading day after formation and held until the next rebalance, with weights drifting in between. The long-short portfolio is long the top and short the bottom tercile, so it is self-financing and its return is already an excess return. The long-only portfolio holds the top tercile and is compared with the DAX.

Costs. Turnover is measured against drifted weights. Every euro traded costs 10 bps, a realistic all-in figure (spread plus fees) for DAX stocks; the short leg pays 25 bps per year. The break-even cost is the one-way cost that would reduce the mean excess return to zero.

Inference. Means and CAPM alphas are tested with Newey–West standard errors. The rank information coefficient (IC) is the Spearman correlation between the signal and the next holding-period return. The probabilistic Sharpe ratio (PSR) is the probability that the true Sharpe ratio exceeds zero given the sample length, skewness and kurtosis. The deflated Sharpe ratio (DSR) replaces zero with the Sharpe ratio one would expect from the best of 36 unskilled trials, here 0.43.

5 Results

The strongest signal is *Momentum 12-1*, whose long-short portfolio earns 7.2% per year after costs (Sharpe 0.40); its mean return is only marginally significant (10% level). The weakest is *Short-term reversal* with a net Sharpe ratio of −0.29; it loses money mostly through trading costs, since its gross spread is only −0.4% per year at a turnover of 16.3 times the portfolio per year. After a Holm correction for testing six strategies, no long-short premium remains significant at the 5% level.

Table 4: Do the signals predict returns?

Signal	IC	t(IC)	IC>0	ICIR	Spread	L/S net	t	Holm p	PSR	DSR	Monot.
Momentum 12-1	0.039	2.31	58%	0.47	8.2%	7.2%*	1.82	0.42	0.97	0.45	yes
Short-term reversal	−0.000	−0.03	49%	−0.01	−0.4%	−4.2%	−1.40	0.81	0.08	0.00	no
Low volatility	0.016	0.81	52%	0.18	2.0%	0.7%	0.17	1.00	0.57	0.04	yes
Low beta	0.019	0.92	51%	0.19	1.7%	0.5%	0.11	1.00	0.55	0.03	no
52-week high	0.022	1.37	54%	0.27	5.1%	3.0%	0.82	1.00	0.78	0.11	yes
Composite	0.028	1.41	52%	0.29	3.0%	1.5%	0.35	1.00	0.64	0.06	yes

IC = mean monthly rank IC, ICIR = annualised IC/standard deviation. Spread = top minus bottom tercile before costs, per year. L/S net = mean long-short return after costs and lending fees, per year; stars mark Newey–West significance at 10, 5 and 1%. Holm p adjusts for six tests. PSR and DSR as defined above. Monot. = average tercile returns decline from top to bottom.

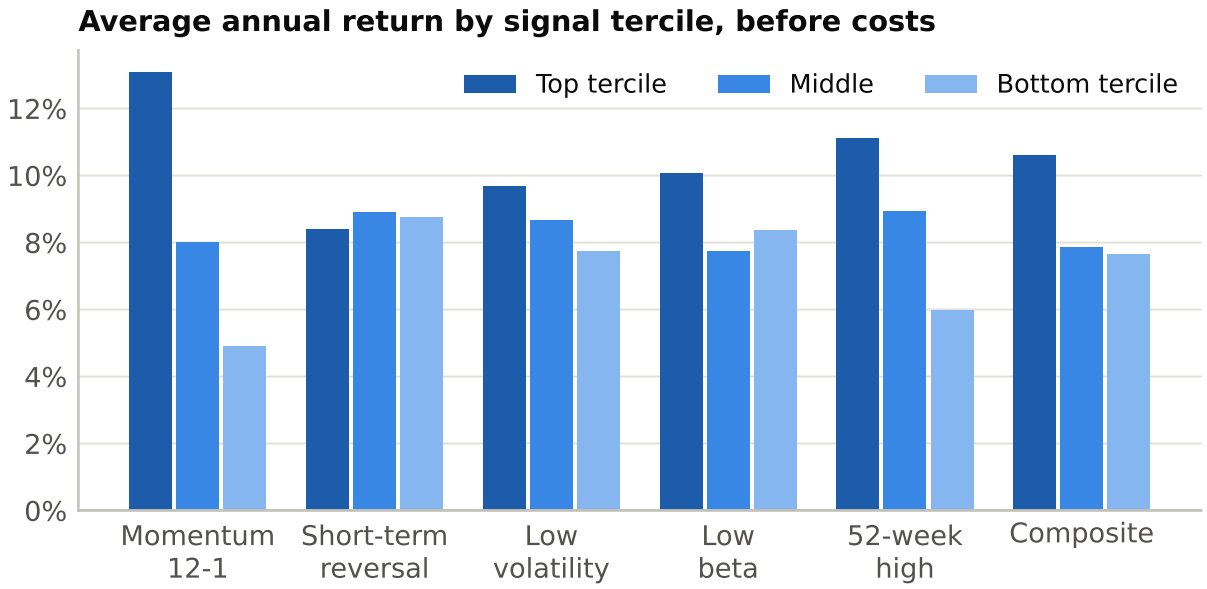


Figure 8: Average annual return of each tercile. A useful signal shows a staircase from left (top) to right (bottom).

Market exposure. Five of six long-short portfolios carry a negative market beta (lowest: Low beta at −0.76), because the bottom tercile of these signals is dominated by volatile, high-beta stocks. In a rising market that drag hides part of the premium: after controlling for the DAX, the CAPM alpha is significant at 5% for Momentum 12-1, Low beta and 52-week high. Table 5 shows the trade-off between gross premium, turnover and cost.

Long only. Most investors cannot short. Table 6 evaluates the top tercile alone against the DAX. The long-only composite portfolio returns 9.8% per year against 7.4% for the DAX, with a beta of 0.65, a tracking error of 9.7% and an information ratio of 0.16. Its CAPM alpha of 4.2% per year has a t-statistic of 2.80.

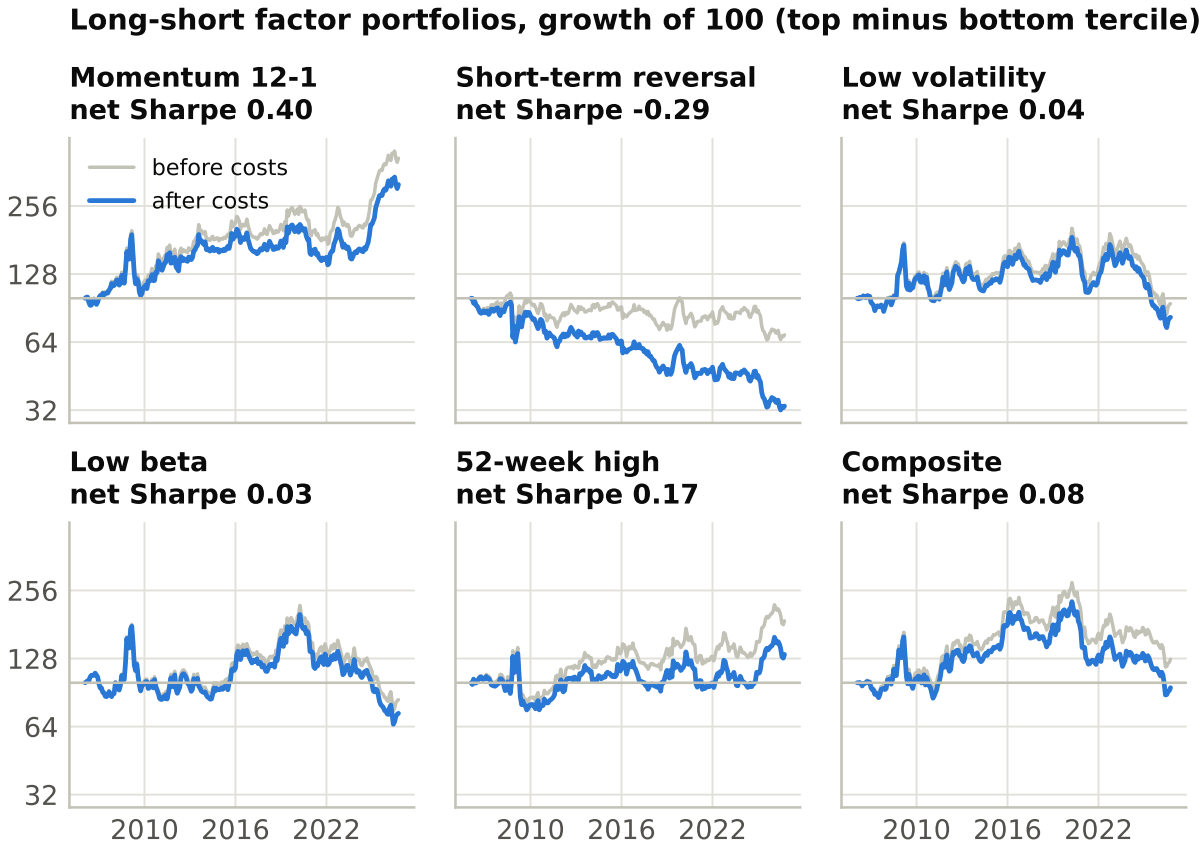


Figure 9: Long-short portfolios, growth of 100 before and after costs.

Table 5: Long-short portfolios after costs

Signal	CAGR	Vol.	Sharpe	Max DD	Beta	Alpha	t	Turnover	Break-even
Momentum 12-1	5.8%	17.9%	0.40	−46.3%	−0.40	10.2%***	2.82	5.3	81
Short-term reversal	−5.2%	14.4%	−0.29	−67.9%	0.12	−5.2%	−1.61	16.3	−2
Low volatility	−0.9%	18.4%	0.04	−60.2%	−0.61	5.4%	1.64	2.1	33
Low beta	−1.5%	20.3%	0.03	−67.3%	−0.76	6.3%**	2.02	2.1	28
52-week high	1.4%	17.9%	0.17	−43.7%	−0.53	7.0%**	2.25	7.1	33
Composite	−0.2%	18.6%	0.08	−61.4%	−0.64	6.3%*	1.95	5.8	25

Beta and alpha from a regression of monthly returns on DAX excess returns, alpha per year with Newey–West t-statistic. Turnover = one-way turnover per year (1.0 means the whole portfolio is replaced once). Break-even in bps per side.

Table 6: Long-only top-tercile portfolios against the DAX

Portfolio	CAGR	Vol.	Sharpe	Max DD	Beta	Alpha	t	TE	IR	Up	Down	Turnover
Momentum 12-1	12.0%	16.3%	0.71	−34.4%	0.79	5.5%***	3.23	9.1%	0.43	94%	71%	2.7
Short-term reversal	4.2%	20.4%	0.25	−60.4%	1.03	−2.7%	−1.32	9.2%	−0.27	96%	110%	8.3
Low volatility	8.7%	13.2%	0.61	−32.9%	0.65	3.2%**	2.39	9.2%	0.06	75%	61%	1.0
Low beta	9.4%	12.7%	0.67	−26.5%	0.56	4.3%***	2.91	11.1%	0.09	68%	49%	1.0
52-week high	9.5%	14.1%	0.63	−39.7%	0.69	3.7%**	2.56	8.9%	0.15	81%	65%	4.0
Composite	9.8%	13.7%	0.67	−37.7%	0.65	4.2%***	2.80	9.7%	0.16	76%	57%	2.9
Equal-weight PIT	7.1%	17.5%	0.41	−53.3%	0.97	−0.1%	−0.11	3.4%	−0.09	97%	97%	0.4

After costs. TE = tracking error, IR = information ratio against the DAX, Up/Down = up- and down-market capture ratios. The equal-weight PIT portfolio shows how much of any outperformance comes from equal weighting alone.

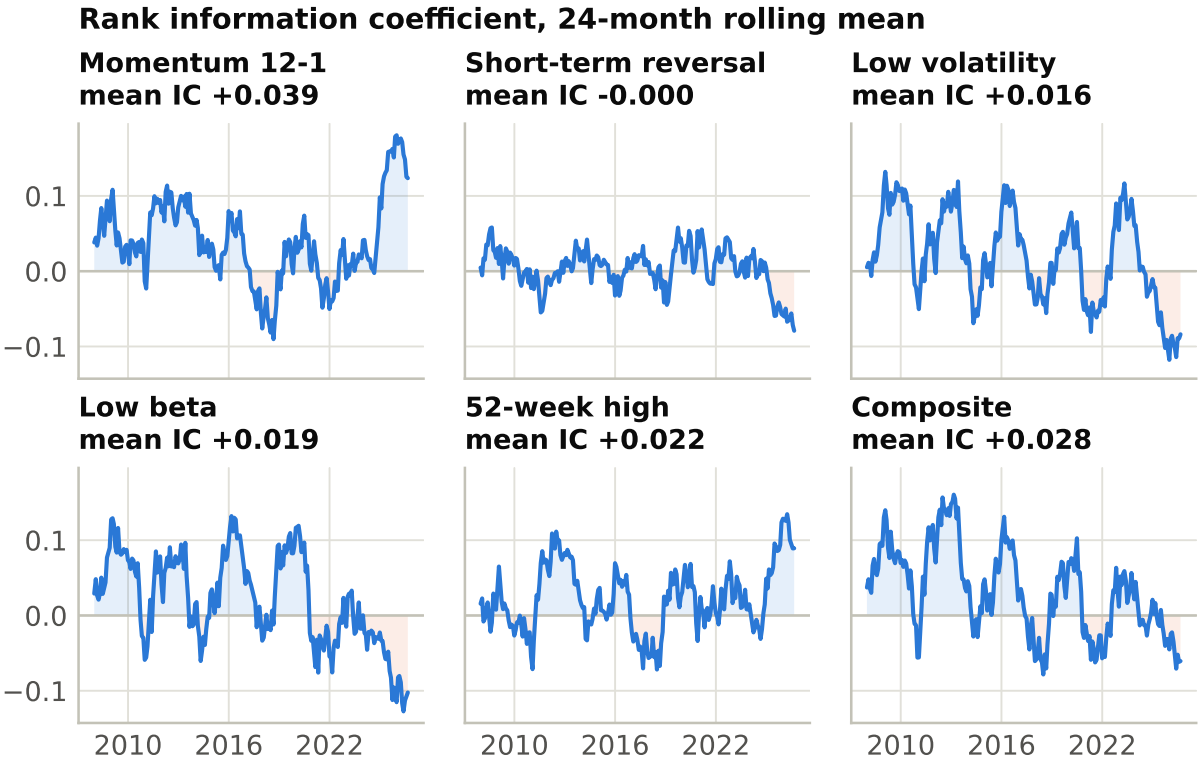


Figure 10: Rolling two-year average of the monthly rank IC. Long stretches above or below zero show that factor premia come and go.

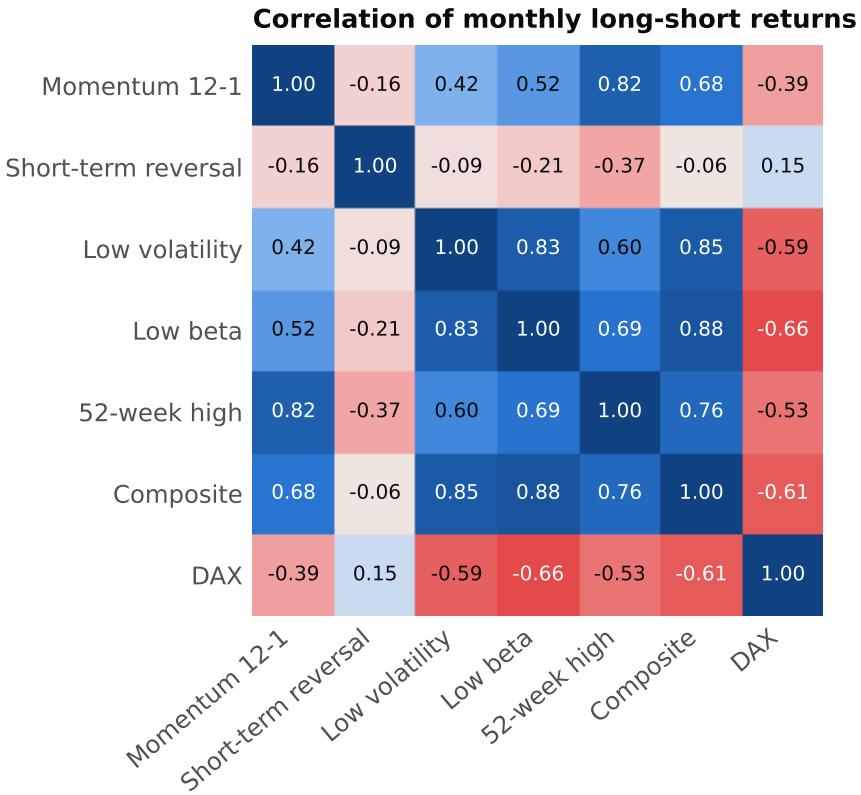


Figure 11: Correlations of monthly long-short returns and the DAX. Highly correlated signals add little diversification when they are combined in the composite.

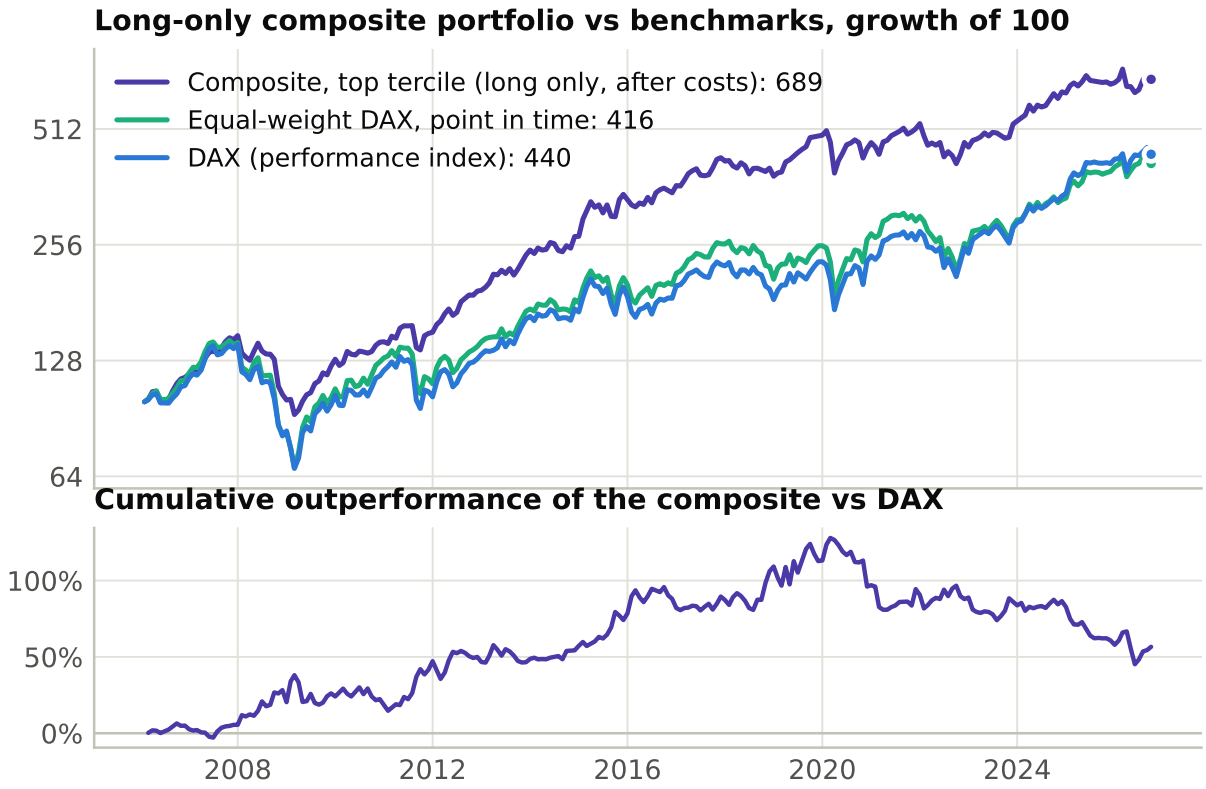


Figure 12: Long-only composite against both benchmarks.

6 Robustness

Figure 13 repeats every long-short strategy with no, one and five days between signal and trade and with terciles or quintiles. All 36 cells enter the deflated Sharpe ratio, so trying more variants makes the bar for significance higher, not lower. Figure 14 shows how quickly high-turnover signals die when costs rise, and Table 7 splits the sample into three regimes.

Table 7: Subperiods: Sharpe ratio / CAGR

Series	Crisis years 2006-2012	Low-rate era 2013-2019	2020 to date
DAX	0.19 / 4.2%	0.62 / 8.2%	0.55 / 10.0%
S&P 500 (EUR)	0.06 / 2.1%	1.41 / 17.8%	0.88 / 15.1%
Equal-weight PIT	0.25 / 5.3%	0.66 / 8.6%	0.42 / 7.6%
Momentum 12-1 (L/S)	0.37 / 6.0%	0.35 / 4.3%	0.52 / 7.1%
Short-term reversal (L/S)	−0.20 / −5.3%	−0.13 / −2.1%	−0.59 / −8.1%
Low volatility (L/S)	0.24 / 2.8%	0.30 / 3.6%	−0.45 / −8.9%
Low beta (L/S)	0.09 / −0.8%	0.54 / 8.2%	−0.59 / −11.3%
52-week high (L/S)	0.14 / 0.6%	0.17 / 1.4%	0.22 / 2.2%
Composite (L/S)	0.26 / 3.4%	0.47 / 6.7%	−0.56 / −10.3%
Composite (long only)	0.59 / 10.1%	1.20 / 14.2%	0.31 / 5.1%

Long-short Sharpe ratios are self-financing; long-only and benchmark Sharpe ratios are in excess of cash.

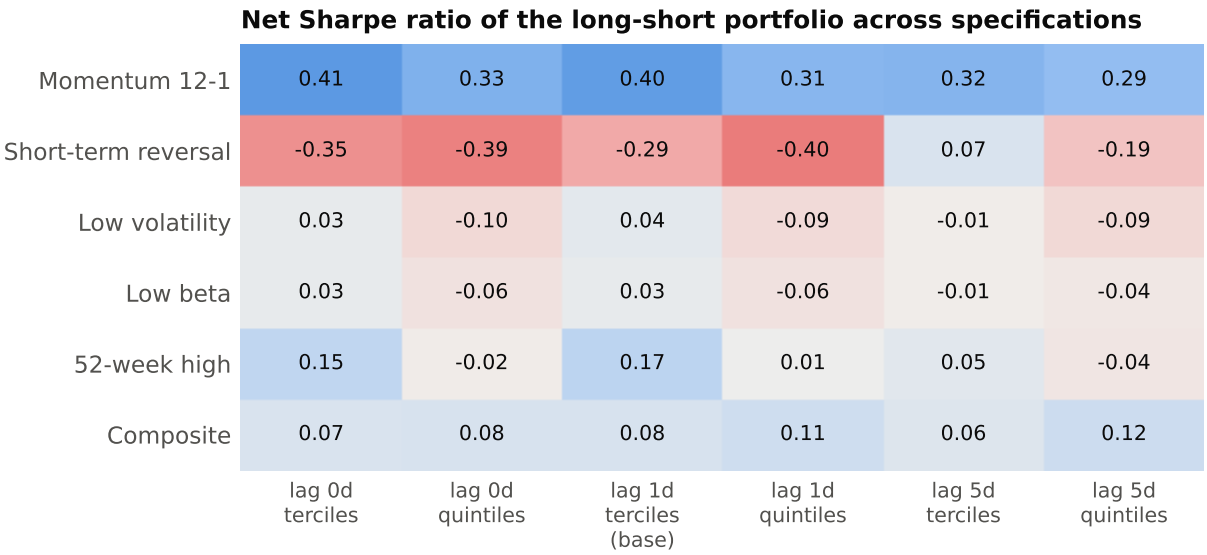


Figure 13: Net Sharpe ratio for each specification. The base case is lag 1, terciles.

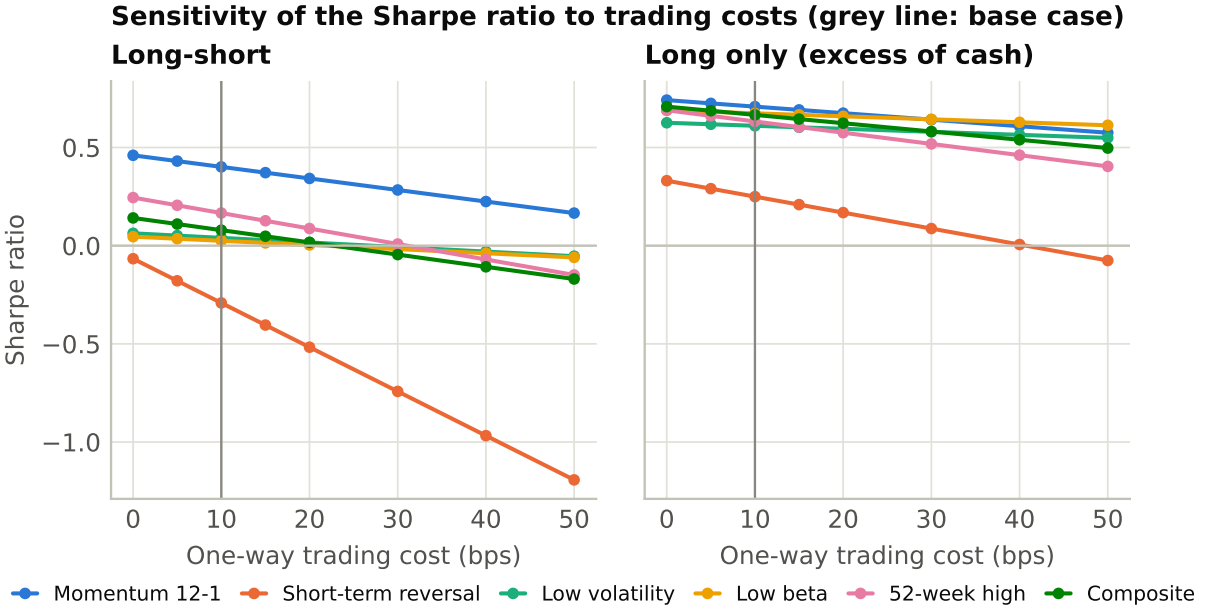


Figure 14: Sharpe ratio as a function of the one-way trading cost.

7 Live out-of-sample record

The specification in `config.py` was frozen on 2026-10-14. Every observation after that date was unknown when the study was designed, so this table is the only clean test. It grows with each data update.

No out-of-sample data yet. The specification was frozen on 2026-10-14; this table fills in automatically with every update.

8 Conclusion

Inside the DAX, simple price-based factors behave much like the literature predicts in direction, but the cross-section is narrow and the evidence is weak once costs, market exposure and multiple testing are taken seriously. After a Holm correction for testing six strategies, no long-short premium remains significant at the 5% level. Across all 36 specifications that were run, an unskilled researcher would expect a best Sharpe ratio of 0.43 by luck alone. No strategy has a deflated Sharpe ratio above 0.95, so none of them can be called a discovery on this data. The defensive signals (low volatility and low beta) give long-only portfolios with an average volatility of 13.0% against 17.7% for the DAX and an average Sharpe ratio of 0.64, while their long-short spreads average a Sharpe ratio of 0.03: in raw returns their benefit is lower risk rather than a dollar-neutral premium. Two lessons reach beyond factor premia: how much a back-filled universe flatters every backtest, and how different the DAX and the S&P 500 have been for a euro investor. The live record will show whether any of the in-sample patterns persist.

Limitations. Free data only; 7 delisted members are missing; prices are adjusted closes, so dividends enter the 52-week high; costs are a single flat estimate; no fundamental factors (value, quality), because point-in-time fundamentals are not freely available.

A Calendar-year returns

Year	DAX	S&P 500 (EUR)	Equal-weight	Composite (long)	Momentum (L/S)
2006	15.2%	2.9%	18.8%	18.2%	0.7%
2007	22.3%	−4.7%	20.1%	25.7%	13.1%
2008	−40.4%	−35.8%	−41.6%	−32.0%	30.9%
2009	23.8%	27.7%	29.8%	27.7%	−24.6%
2010	16.1%	23.6%	20.4%	11.0%	26.3%
2011	−14.7%	4.2%	−14.4%	5.8%	7.1%
2012	29.1%	11.5%	28.6%	28.6%	−1.8%
2013	25.5%	29.2%	21.6%	27.2%	19.1%
2014	2.7%	30.5%	3.9%	8.6%	−8.1%
2015	9.6%	12.7%	11.3%	24.4%	15.2%
2016	6.9%	14.6%	7.2%	9.0%	−11.7%
2017	12.5%	7.9%	19.0%	15.8%	−0.2%
2018	−18.3%	−1.0%	−19.0%	−8.7%	8.1%
2019	25.5%	35.2%	22.3%	27.8%	11.5%
2020	3.5%	7.7%	7.5%	−4.3%	−22.0%
2021	15.8%	40.1%	11.0%	12.0%	−2.6%
2022	−12.3%	−13.0%	−16.2%	−13.1%	17.8%
2023	20.3%	21.6%	16.8%	17.0%	−11.2%
2024	18.8%	33.2%	13.9%	18.1%	33.8%
2025	23.0%	4.8%	20.5%	6.3%	47.0%
2026	2.9%	16.2%	2.0%	2.0%	2.0%

B Every constituent

Statistics only cover the days a stock was in the index. † = no longer a member.

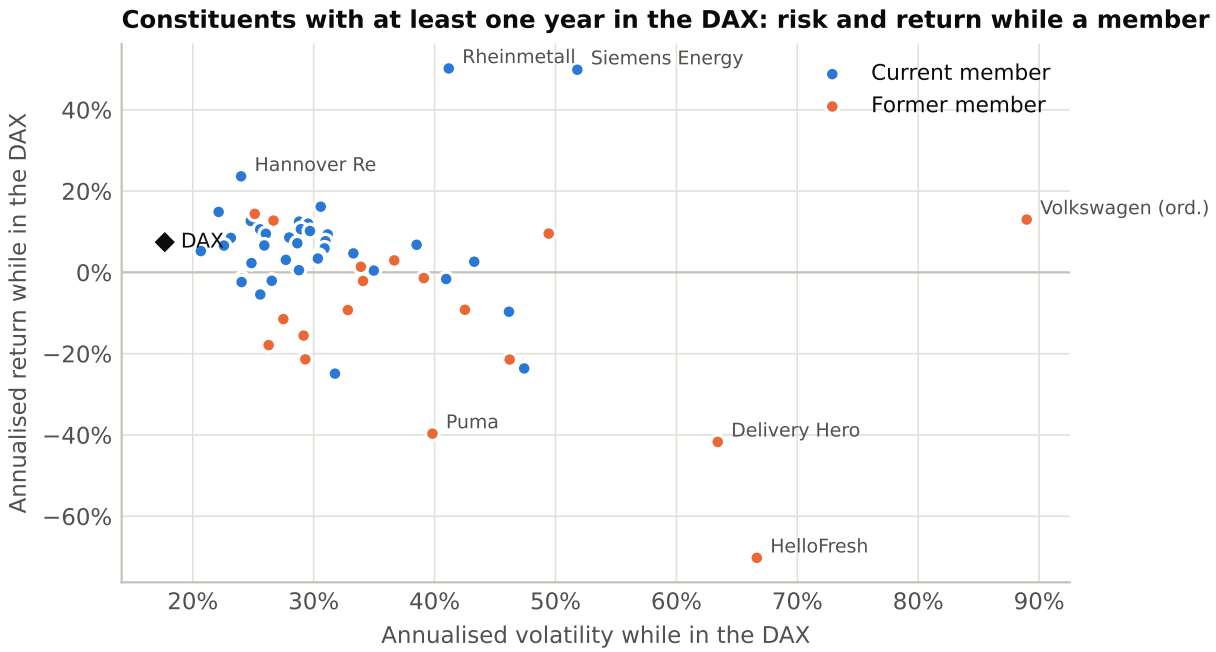


Figure 15: Annualised return and volatility of each stock while it was a DAX member, 2006-02 onwards. Stocks with less than one year of membership are left out of the chart but are in the table.

Company	Sector	Years	Return	Vol.	Sharpe	Beta	Max DD	Worst day
Rheinmetall	Industrials	3.6	50.2%	41.2%	1.12	1.02	−53.0%	−18.7%
Siemens Energy	Industrials	5.1	49.8%	51.8%	1.02	1.48	−71.9%	−37.3%
Hannover Re	Financials	5.1	23.6%	24.0%	0.91	0.65	−17.5%	−7.7%
Daimler Truck	Industrials	4.6	16.2%	30.6%	0.56	1.14	−34.6%	−8.7%
GEA Group	Industrials	1.1	14.9%	22.1%	0.64	0.68	−16.7%	−5.0%
Linde†	Materials	17.2	14.4%	25.1%	0.63	0.44	−38.2%	−9.5%
Volkswagen (ord.)†	Consumer	3.9	13.0%	89.0%	0.40	0.45	−85.7%	−22.7%
Deutsche Wohnen†	Real Estate	1.4	12.7%	26.7%	0.60	0.33	−17.5%	−8.2%
Munich Re	Financials	20.8	12.6%	24.8%	0.56	0.84	−48.4%	−17.7%
Airbus	Industrials	5.1	12.5%	28.8%	0.48	1.22	−27.7%	−11.6%
Deutsche Börse	Financials	20.8	12.0%	29.5%	0.49	0.84	−76.9%	−12.1%
Allianz	Financials	20.8	10.6%	28.9%	0.45	1.10	−72.7%	−15.3%
Fresenius	Health Care	17.7	10.6%	25.6%	0.49	0.68	−72.3%	−17.7%
Siemens	Industrials	20.8	10.2%	29.7%	0.43	1.15	−67.3%	−17.1%
Salzgitter†	Materials	1.5	9.5%	49.4%	0.41	1.50	−39.9%	−16.1%
SAP	Technology	20.8	9.5%	26.0%	0.43	0.81	−53.0%	−21.9%
Heidelberg Materials	Materials	16.4	9.3%	31.2%	0.42	1.18	−66.6%	−14.2%
DHL Group	Industrials	20.8	8.6%	28.0%	0.39	0.94	−72.1%	−17.3%
Deutsche Telekom	Telecom	20.8	8.5%	23.2%	0.42	0.69	−40.6%	−13.0%
Adidas	Consumer	20.8	7.7%	31.0%	0.35	0.90	−71.5%	−15.4%
BASF	Materials	20.8	7.2%	28.6%	0.34	1.08	−60.3%	−13.6%
MTU Aero Engines	Industrials	7.1	6.8%	38.5%	0.33	1.22	−62.9%	−15.6%
Merck KGaA	Health Care	19.5	6.6%	25.9%	0.34	0.63	−52.6%	−14.7%
Henkel	Consumer	20.8	6.6%	22.6%	0.34	0.62	−56.3%	−11.3%
BMW	Consumer	20.8	5.9%	30.9%	0.30	1.11	−65.4%	−13.2%
Beiersdorf	Consumer	17.1	5.2%	20.7%	0.31	0.49	−53.3%	−20.1%
Mercedes-Benz Group	Consumer	20.8	5.2%	33.5%	0.28	1.27	−76.5%	−18.9%
Volkswagen (pref.)	Consumer	16.9	4.7%	33.3%	0.28	1.23	−63.0%	−19.8%
Bayer	Health Care	20.8	4.1%	30.4%	0.24	0.93	−82.3%	−18.0%
RWE	Utilities	20.8	3.4%	30.3%	0.22	0.81	−85.4%	−17.3%
E.ON	Utilities	20.8	3.1%	27.7%	0.20	0.82	−77.3%	−11.5%
Metro (old)†	Consumer	6.7	2.9%	36.7%	0.20	0.80	−71.5%	−13.9%
Infineon	Technology	20.3	2.6%	43.3%	0.25	1.26	−97.1%	−39.6%
Fresenius Medical Care	Health Care	19.1	2.3%	24.9%	0.18	0.49	−69.1%	−16.5%
Lufthansa†	Industrials	14.5	1.4%	33.9%	0.18	0.97	−75.2%	−14.2%
Vonovia	Real Estate	11.1	0.5%	28.8%	0.13	0.65	−65.8%	−12.1%
Continental	Consumer	17.0	0.4%	35.0%	0.15	1.12	−76.7%	−17.5%
thyssenkrupp†	Materials	13.7	−1.4%	39.1%	0.13	1.31	−74.7%	−12.5%
Qiagen	Health Care	5.1	−1.6%	24.1%	−0.04	0.42	−43.4%	−10.8%
Deutsche Bank	Financials	20.8	−1.6%	41.0%	0.13	1.43	−92.7%	−18.4%
Brenntag	Materials	5.1	−2.1%	26.5%	−0.03	0.82	−44.7%	−9.6%

Company	Sector	Years	Return	Vol.	Sharpe	Beta	Max DD	Worst day
Covestro†	Materials	6.8	−2.1%	34.1%	0.08	1.04	−69.1%	−14.6%
Symrise	Materials	5.1	−2.4%	24.0%	−0.07	0.47	−47.0%	−9.1%
Siemens Healthineers	Health Care	5.1	−5.4%	25.6%	−0.18	0.78	−45.4%	−8.6%
K+S†	Materials	7.5	−9.2%	42.5%	−0.03	0.99	−70.1%	−24.7%
TUI†	Consumer	2.7	−9.3%	32.8%	−0.25	1.09	−50.5%	−8.5%
Commerzbank	Financials	16.4	−9.7%	46.2%	−0.02	1.37	−98.1%	−24.0%
Lanxess†	Materials	3.0	−11.5%	27.5%	−0.31	0.90	−49.3%	−7.0%
Porsche SE†	Financials	4.8	−15.6%	29.2%	−0.51	1.05	−62.1%	−10.9%
ProSiebenSat.1 Media†	Telecom	2.0	−17.9%	26.3%	−0.60	0.93	−40.7%	−14.5%
Porsche AG†	Consumer	2.8	−21.4%	29.3%	−0.78	0.91	−63.9%	−7.2%
Sartorius†	Health Care	4.1	−21.5%	46.2%	−0.34	1.15	−70.7%	−18.5%
Zalando	Consumer	5.1	−23.6%	47.4%	−0.38	1.37	−82.0%	−13.4%
Scout24	Technology	1.1	−24.9%	31.8%	−0.81	0.48	−40.7%	−6.1%
Puma†	Consumer	1.3	−39.7%	39.8%	−1.07	1.28	−61.3%	−9.0%
Delivery Hero†	Consumer	1.8	−41.7%	63.4%	−0.52	1.23	−82.5%	−30.4%
Hochtief	Industrials	0.3	−58.7%	33.8%	−2.51	1.26	−28.4%	−9.4%
HelloFresh†	Consumer	1.0	−70.2%	66.6%	−1.48	1.41	−75.6%	−14.0%

References

Ang, A., Hodrick, R., Xing, Y., Zhang, X. (2006). The cross-section of volatility and expected returns. *Journal of Finance* 61(1), 259–299.

Bailey, D., López de Prado, M. (2014). The deflated Sharpe ratio. *Journal of Portfolio Management* 40(5), 94–107.

Blitz, D., van Vliet, P. (2007). The volatility effect. *Journal of Portfolio Management* 34(1), 102–113.

Frazzini, A., Pedersen, L. (2014). Betting against beta. *Journal of Financial Economics* 111(1), 1–25.

George, T., Hwang, C. (2004). The 52-week high and momentum investing. *Journal of Finance* 59(5), 2145–2176.

Harvey, C., Liu, Y., Zhu, H. (2016). ... and the cross-section of expected returns. *Review of Financial Studies* 29(1), 5–68.

Jegadeesh, N. (1990). Evidence of predictable behavior of security returns. *Journal of Finance* 45(3), 881–898.

Jegadeesh, N., Titman, S. (1993). Returns to buying winners and selling losers. *Journal of Finance* 48(1), 65–91.

Mertens, E. (2002). Comments on variance of the IID estimator in Lo (2002). Working paper, University of Basel.